

Qazi Noorul Mateen

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RESEARCH INTERESTS

Deep learning for medical imaging and diagnostics; clinical NLP and large language models for healthcare; point-of-care diagnostic systems; trustworthy and privacy-preserving medical AI. I combine a biomedical research foundation (published in *Science Advances*) with hands-on experience designing and deploying production healthcare-AI systems, and am seeking a PhD at the intersection of machine learning and medicine.

EDUCATION

Liverpool John Moores University, Liverpool, UK Jun 2025
MS, Data Science. Thesis: Machine-learning models on speech data to predict Parkinson's disease.

Indian Institute of Technology Delhi, Delhi, India May 2024
MS (Research), Biochemical Engineering & Biotechnology. Thesis: Early gene-expression changes associated with NAFLD toward candidate gene prioritization (Prof. Ishaan Gupta). Included one year as a Graduate Student Researcher at Harvard University.

International Institute of Information Technology, Bangalore (Remote) Apr 2022
PG Diploma, Machine Learning & AI. Capstone: GAN (modified U-Net) for synthetic MRI generation.

Aligarh Muslim University, Aligarh, India — *MSc, Biotechnology*. 2020

Jamia Millia Islamia, New Delhi, India — *BSc, Biosciences*. 2018

RESEARCH & PROFESSIONAL EXPERIENCE

MarkiTech.AI, Oakville, ON, Canada Jun 2025 – Present
AI/ML Specialist

- Lead the design and development of **Heyliaa**, a healthcare AI-agent platform automating clinic front-desk operations — call handling, appointment scheduling, patient check-in, reminders, and fax/insurance workflows — piloting across multiple cardiology clinic sites (~200 patients/day) and two EMR systems (Accuro, Jane).
- Built **LLM-powered clinical voice and messaging agents** integrating speech recognition, large language models, and telephony (Twilio), with live dashboards surfacing real-time transcripts and follow-up actions.
- Architected a cross-EMR **robotic process automation (RPA)** layer (FastAPI + Playwright, Dockerized on Google Cloud) that automates booking, rescheduling, and check-in against EMRs without public APIs, using an asynchronous task-queue pattern (Firestore queue, status polling, webhook callbacks).
- Engineered a **HIPAA-grade security and compliance framework**: AES-256-GCM PHI encryption at rest, SHA-256-hashed patient identifiers, immutable audit logging, PHI redaction in logs, MFA, and session-timeout policies.
- Fine-tuned GPT-2 and larger language models** to generate clinical SOAP notes from therapy transcripts — building the dataset pipeline, training, and a ROUGE + human-rating evaluation framework.

Vgenomics (Remote) Jul 2024 – Present
AI Strategy Advisor

- Advise on AI-platform development for rare-disease diagnostics and genomics, focused on biomarker discovery, technical feasibility, and clinical alignment.

Harvard University / Brigham & Women's Hospital, Harvard Medical School, MA, USA Dec 2022 – Nov 2023

Graduate Student Researcher (J1 Exchange Program) — Supervisor: Prof. Hadi Shafiee; direct supervision: Dr. Hui Chen.

- **Deep-learning point-of-care HCV diagnostics:** Built the deep-learning core of a smartphone-based microfluidic system for rapid HCV antigen detection, applying adaptive adversarial neural networks to 982 unannotated and 1000 annotated images and reaching >94% diagnostic accuracy in clinical validation; designed to reduce diagnostic disparities in underserved communities. *Published in Science Advances (2025).*
- **Fentanyl detection:** Developed a fentanyl-detection assay using microfluidic bubbling technology with a deep-learning readout; presented at the MIT–MGH AI Cures conference, Cambridge, USA.

Indian Institute of Technology Delhi, Delhi, India

Sep 2020 – May 2024

Graduate Researcher & Teaching Assistant

- Developed automated pipelines for differential gene expression and WGCNA on RNA-seq data to prioritize candidate biomarkers for early NAFLD detection.
- Built extraction/classification algorithms to structure unstructured clinical text from patient records for downstream analysis.
- Teaching Assistant, BBL735: Genomics and Proteomics — bioinformatics and ML-in-genomics training.

CSIR–NIEST & National Centre for Biological Sciences, Bangalore, India

Jun – Aug 2020

Summer Research Fellow

- DAE-funded project: in-silico characterization of Domains of Unknown Function (DUF) across bacterial genomes (Dr. Ashwin Sai Narain Seshasayee).

PUBLICATIONS

- Chen H., **Mateen Q.N.**, et al. (2025). Reducing hepatitis C diagnostic disparities with a fully automated deep-learning-enabled microfluidic system for HCV antigen detection. **Science Advances**. doi:10.1126/sciadv.adt3803
- Raghav A., **Mateen Q.N.**, et al. (2021). Mesenchymal stromal cell-derived tailored exosomes treat bacteria-associated diabetic foot ulcers. **Frontiers in Microbiology**.
- Raghav A., **Mateen Q.N.**, et al. (2021). Newer horizon of mesenchymal stem cell-based therapy in SARS-CoV-2-associated mucormycosis. **Frontiers in Microbiology**, 12:738983.
- Masroor A., **Mateen Q.N.**, et al. (2021). Evaluation of ThT augmentation and RLS inner-filter effect as a true inhibitor of amyloid fibrillation. **Archives of Biochemistry and Biophysics**, 709:108981.

SELECTED MACHINE-LEARNING PROJECTS

- **Clinical SOAP-note generation (LLMs):** Fine-tuned GPT-2 and 7B-parameter models to draft SOAP notes from therapy transcripts; built dataset generation, training, and ROUGE + human evaluation.
- **Generative Medical Imaging (CycleGAN / U-Net):** T1↔T2 MRI translation with custom loss functions for synthetic medical-image generation.
- **Parkinson's detection from speech:** Comparative ML study on voice biomarkers (MS thesis, LJMU).
- **Medical Named-Entity Recognition:** CRF-based NER (spaCy, scikit-learn) extracting diseases, treatments, and symptoms from biomedical text.

TECHNICAL SKILLS

Machine Learning / DL: PyTorch, TensorFlow, scikit-learn, Hugging Face; CNN, RNN, GAN, adversarial learning; NLP, LLMs, fine-tuning; AI agents / tool-calling.

Medical & Health AI: Medical image analysis, generative medical imaging, point-of-care diagnostics, clinical NLP / NER, biomarker discovery, HIPAA-grade privacy engineering.

Programming & Cloud/MLOps: Python, R, JavaScript, Bash; Google Cloud (Cloud Run, Firestore), AWS, Docker, Terraform, HPC.

Computational Biology: RNA-seq (DESeq2, edgeR, Bioconductor), WGCNA, single-cell pipelines.

Biomedical Wet Lab: Microfluidic device fabrication, 3D printing, PCR, ELISA, DNA/RNA extraction, CRISPR-Cas, nanopore sequencing, BSL2.

CONFERENCES & CERTIFICATIONS

Poster: Deep-learning–assisted detection of fentanyl using a bubbling-microchip — MIT–MGH AI Cures, Cambridge, USA. · RECOMB 2020 (Computational Molecular Biology), Padova, Italy. · Google Cloud Essentials · Career Essentials in Data Analysis (Microsoft & LinkedIn) · GIAN course, Laval University (10/10).

HONORS & AWARDS

GATE 2020 — All India Rank 161 of 20,646 · DBT Biotech Fellowship — All India Rank 3 · DBT–India Alliance Research Fellowship — top 40 of 14,000.